

# Exact Tradeoff Between Quantum Error Correction and Quantum Darwinism: An Information-Theoretic No-Go Theorem

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Quantum error correction (QEC) and Quantum Darwinism describe opposing consequences of system–environment interactions: QEC seeks to preserve logical quantum information, whereas Quantum Darwinism explains the emergence of objective classical information through its proliferation into the environment. Despite their common physical origin, no direct quantitative connection between these paradigms has previously been established. We introduce an exactly solvable block–environment model based on the logical GHZ block of the Shor  $[[9,1,3]]$  quantum error-correcting code, collectively coupled to  $N$  environment qubits. The logical fidelity, Holevo information, and Darwinistic redundancy are obtained systematically for arbitrary environment size and imperfect recovery efficiency. Eliminating the common decoherence parameter yields an exact tradeoff relating Darwinistic redundancy directly to the post-recovery logical fidelity, demonstrating that the emergence of redundant classical records occurs at the expense of logical quantum information. We further prove a model-independent no-go theorem showing that the logical fidelity exceeds a critical threshold precludes the emergence of Darwinistic redundancy, irrespective of the microscopic Hamiltonian or environment structure. The solvable model saturates this general bound, establishing the first quantitative information-theoretic connection between logical quantum information protection and the emergence of redundant classical records.

**Introduction:** Quantum information can either remain protected within a quantum system or become encoded in its surrounding environment, but it cannot do both simultaneously. This fundamental competition lies at the heart of two major developments in quantum information science: quantum error correction [1–7], which protects quantum coherence against environmental noise, and quantum Darwinism [8–14], which explains the emergence of objective classical reality through the redundant encoding of information in the environment. Despite their common origin in system–environment interactions, a direct quantitative connection between these two paradigms has not been established. Such a connection would reveal the fundamental limits governing both the protection of quantum coherence and the formation of classical objectivity, and would place the quantum-to-classical transition on quantitative footing.

The absence of such a quantitative connection is closely related to the lack of a common analytical framework in which both quantum error correction and Quantum Darwinism can be treated on equal footing. While QEC is naturally characterized by logical fidelities after recovery, Quantum Darwinism is quantified through the Holevo information and the redundancy of environment fragments. Bridging these paradigms therefore requires a model in which both quantities can be obtained analytically from the same microscopic dynamics. A very recent study explored connections between Quantum Darwinism and quantum error correction from an operator-algebraic perspective [15]. Complementary to this algebraic approach, we develop an exactly solvable dynamical model in which both the logical fidelity and the Darwinistic redundancy are obtained in closed form. Eliminating the common decoherence parameter yields an analytical tradeoff that is independent of the microscopic dynamical parameters.

In this work, we introduce a block–environment model in which a logical qubit is encoded in one GHZ block of the Shor  $[[9, 1, 3]]$  code and interacts collectively with  $N$  independent environment qubits through a non-commuting Hamiltonian. Because each GHZ block experiences identical block–

environment dynamics in the present model, analyzing a single block faithfully captures the information flow throughout the full nine-qubit encoding. The commuting sector of the model is exactly solvable, yielding closed-form expressions for the logical fidelity, Holevo information, and Darwinistic redundancy for arbitrary environment size and imperfect recovery efficiency. The robustness of these analytical results against the non-commuting interaction is established by exact numerical simulations.

Our central result is a closed-form tradeoff theorem showing that the Darwinistic redundancy  $R_\delta$  is uniquely determined by the post-recovery logical fidelity  $F_L^{(N)}$ , with the microscopic dynamical parameters eliminated from the final relation. Any gain in classical objectification therefore comes at a quantifiable cost to quantum error-correction performance. Beyond the solvable model, we prove a model-independent no-go theorem showing that whenever the bare logical fidelity exceeds a threshold set by the binary entropy, Darwinistic redundancy cannot emerge irrespective of the microscopic Hamiltonian or environment structure. Together, these results establish the first quantitative information-theoretic framework linking logical quantum information protection to the emergence of redundant classical information.

These results also have a direct operational consequence: syndrome-based recovery [16–22] and environment-fragment readout [12–14], although experimentally distinct, become equivalent probes of the same underlying decoherence process. Measuring the post-recovery logical fidelity uniquely determines the Darwinistic information encoded in the environment, and vice versa. Because the solvable model saturates the general no-go bound, it represents an extremal realization of the competition between logical quantum information protection and redundant classical information, providing a general framework for exploring the transition from protected logical quantum information to redundant classical information.

**Exactly Solvable Block–Environment Model.**— We consider the logical GHZ block of the Shor  $[[9, 1, 3]]$  quantum

error-correcting code interacting collectively with an environment of  $N$  independent qubits. Since each block–environment pair undergoes identical dynamics, it is sufficient to focus on a representative logical block  $b$ . The logical basis is formed by the orthogonal codewords  $|\bar{z}_\pm\rangle_b = (|000\rangle \pm |111\rangle)/\sqrt{2}$ , while each environment qubit is initially prepared in  $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ . The block interacts with the environment through the Hamiltonian

$$\hat{H} = g_Z \hat{Z}_b \otimes \hat{S}_Z + g_X \hat{X}_b \otimes \hat{S}_X; \quad \hat{S}_Z = \sum_{k=1}^N Z_k, \quad \hat{S}_X = \sum_{k=1}^N X_k, \quad (1)$$

where  $\hat{Z}_b = Z^{\otimes 3}$  and  $\hat{X}_b = X^{\otimes 3}$  act on the three-qubit logical block, while  $\hat{S}_Z$  and  $\hat{S}_X$  act collectively on the  $N$ -qubit environment. Within the logical subspace,  $\hat{Z}_b$  exchanges the two logical codewords ( $\hat{Z}_b |\bar{z}_\pm\rangle = |\bar{z}_\mp\rangle$ ), whereas  $\hat{X}_b$  acts as the corresponding logical phase operator ( $\hat{X}_b |\bar{z}_\pm\rangle = \pm |\bar{z}_\pm\rangle$ ). The analytical results are obtained in the exactly solvable limit  $\hat{H}_0 = g_Z \hat{Z}_b \otimes \hat{S}_Z$ , corresponding to  $g_X = 0$  (or perturbatively  $g_X \ll g_Z$ ). The full Hamiltonian Eq. (1) is retained in the numerical analysis to test the robustness of the analytical tradeoff beyond the exactly solvable regime.

Since  $[\hat{Z}_b \otimes Z_i, \hat{Z}_b \otimes Z_j] = 0$ , the propagator generated by  $H_0$  factorizes exactly,  $\hat{U}_0(t) = e^{-it\hat{H}_0} = \prod_{k=1}^N \exp(-ig_Z t \hat{Z}_b \otimes Z_k)$ . The detailed derivation is given in the Appendix. Starting from  $|\Psi(0)\rangle = |\bar{z}_+\rangle_b \otimes |+\rangle^{\otimes N}$ , the exact evolved state admits the branch decomposition

$$|\Psi_+(t)\rangle = \frac{1}{2}(|\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b)|\phi_+(t)\rangle + \frac{1}{2}(|\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b)|\phi_-(t)\rangle, \quad (2)$$

where the conditional environment states are  $|\phi_\pm(t)\rangle = e^{\mp ig_Z t \hat{S}_Z} |+\rangle^{\otimes N}$ . The analytically solvable interaction is a pure-dephasing model in the  $\hat{Z}_b$  eigenbasis  $\{|000\rangle, |111\rangle\}$ , analogous to standard spin–environment dephasing models [23]. Throughout this work, however, we describe the dynamics in the logical basis  $\{|\bar{z}_+\rangle, |\bar{z}_-\rangle\}$ , where  $\hat{Z}_b$  exchanges the logical codewords ( $\hat{Z}_b |\bar{z}_\pm\rangle = |\bar{z}_\mp\rangle$ ), so that the same interaction appears as logical-state transitions. Because the operators  $Z_k$  commute, these states admit the product representation

$$|\phi_\pm(t)\rangle = \bigotimes_{k=1}^N [\cos(g_Z t)|+\rangle_k \mp i \sin(g_Z t)|-\rangle_k]. \quad (3)$$

Tracing out the environment gives  $\rho_b(t) = \text{diag}[(1+\Gamma_N)/2, (1-\Gamma_N)/2]$  in the logical basis  $\{|\bar{z}_+\rangle_b, |\bar{z}_-\rangle_b\}$ , governed entirely by the decoherence factor

$$\Gamma_N(t) = \langle \phi_-(t) | \phi_+(t) \rangle = \cos^N(2g_Z t). \quad (4)$$

Thus the complete reduced dynamics of the logical block is governed by the single decoherence factor  $\Gamma_N(t)$ .

*Logical fidelity.*— We model recovery via the imperfect channel  $\mathcal{E}_\eta(\rho) = P_+ \rho P_+ + \eta (\hat{Z}_b \otimes \mathbb{I}_E) P_- \rho P_- (\hat{Z}_b \otimes \mathbb{I}_E) + (1-\eta) P_- \rho P_-$ , where  $P_\pm = |\bar{z}_\pm\rangle_b \langle \bar{z}_\pm| \otimes \mathbb{I}_E$  are the syndrome

projectors, and  $\eta \in [0, 1]$  denotes the probability of successfully correcting the  $P_-$  branch. Tracing out the environment after applying  $\mathcal{E}_\eta$  gives the exact post-recovery logical fidelity

$$F_L^{(N)}(t, \eta) = \frac{1 + \eta + (1 - \eta)\Gamma_N(t)}{2}. \quad (5)$$

*Holevo information.*— The classical information accessible to an observer holding a fragment of  $m$  environment qubits is quantified by the Holevo information

$$\chi_m = S(\bar{\rho}_F) - \frac{1}{2}S(\rho_F^+) - \frac{1}{2}S(\rho_F^-), \quad (6)$$

where  $S(\rho) = -\text{Tr}(\rho \ln \rho)$  is the von Neumann entropy,  $\rho_F^\pm$  are the fragment states conditioned on the two logical codewords, and  $\bar{\rho}_F = (\rho_F^+ + \rho_F^-)/2$  is their average. Because  $|\phi_\pm(t)\rangle$  are product states, the conditional fragment states are pure ( $S(\rho_F^\pm) = 0$ ) and their overlap factorizes as  $\gamma_m(t) = \cos^m(2g_Z t) = [\Gamma_N(t)]^{m/N}$ . The Holevo information is therefore (Appendix)

$$\chi_m(t) = H_2\left(\frac{1 + \cos^m(2g_Z t)}{2}\right), \quad (7)$$

where  $H_2(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy. For any nontrivial interaction ( $|\cos(2g_Z t)| < 1$ ), the overlap decreases exponentially with fragment size, so  $\chi_m$  increases monotonically toward its maximum value  $\ln 2$ , reflecting the increasing distinguishability of the conditional fragment states and hence the progressive accumulation of accessible classical information about the logical state in the environment.

*Darwinistic redundancy.*—Following Refs. [9, 10, 24, 25], the Darwinistic redundancy  $R_\delta$  quantifies how many independent observers can simultaneously infer the logical state by accessing disjoint environment fragments. Here  $\delta \in (0, 1)$  specifies the tolerated information deficit relative to the maximum accessible classical information  $\ln 2$ . The redundancy is defined as  $R_\delta = N/m_\delta$ , where  $m_\delta = \min\{m : \chi_m \geq (1 - \delta) \ln 2\}$  is the minimum fragment size satisfying the Darwinism criterion. Combining Eqs. (7) and (4) yields the closed-form expression

$$R_\delta(t) = N \frac{\ln[\cos(2g_Z t)]}{\ln \gamma^*(\delta)}, \quad (8)$$

where  $\gamma^*(\delta) \in (0, 1)$  is the threshold fragment overlap defined implicitly by  $H_2\left(\frac{1+\gamma^*}{2}\right) = (1 - \delta) \ln 2$ . Equation (8) establishes the analytical dependence of Darwinistic redundancy on the microscopic decoherence process, providing the quantitative link between logical quantum information preservation and the redundant encoding of classical information in the environment.

*Exact tradeoff theorem.*— The logical fidelity and the Darwinistic redundancy are both controlled by the same decoherence factor  $\Gamma_N(t)$ . Eliminating  $\Gamma_N$  between Eqs. (5) and (8) yields the central result

$$R_\delta = \frac{1}{\ln \gamma^*(\delta)} \ln \left[ \frac{2F_L^{(N)} - (1 + \eta)}{1 - \eta} \right]. \quad (9)$$

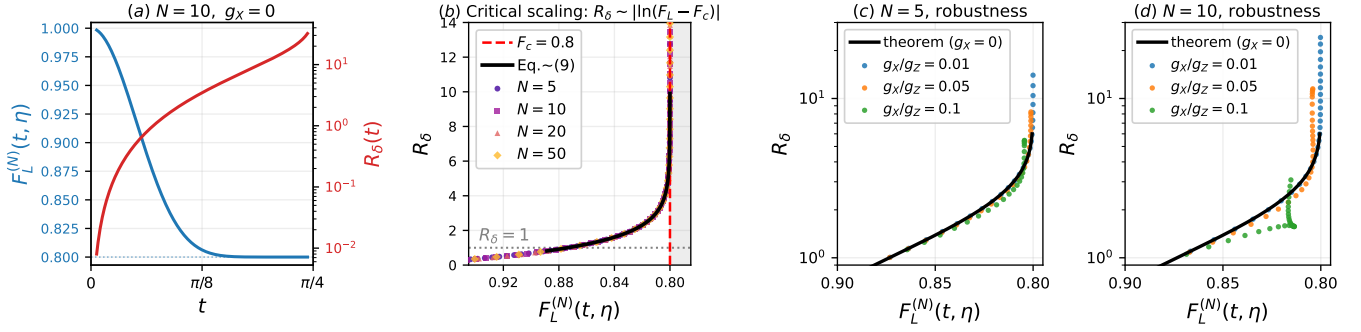


FIG. 1. Numerical results for the Darwinism–QEC competition ( $g_Z = 1$ ,  $\eta = 0.6$ ,  $\delta = 0.10$ ). (a) Competition in time for  $N = 10$ ,  $g_X = 0$ . Blue curve (left axis): logical fidelity  $F_L^{(N)}(t, \eta)$ , decaying monotonically from unity toward the floor  $F_c = (1 + \eta)/2 = 0.8$  (dotted line). Red curve (right axis, log scale): Darwinistic redundancy  $R_\delta(t)$ , rising simultaneously. Solid lines are the closed-form analytical predictions Eqs. (5) and (8); exact diagonalization data (not shown separately) agrees to these. (b) Critical Darwinism–QEC scaling.  $R_\delta$  is plotted against the logical fidelity  $F_L^{(N)}$  on linear axes for  $N = 5, 10, 20, 50$  (colored points). The black curve is the exact tradeoff (Eq. (9)). Data from all four environment sizes collapse onto the same analytical curve, confirming that  $N$  is absorbed entirely into the fidelity gap  $\varepsilon = F_L^{(N)} - F_c$ . The red dashed line marks the critical fidelity  $F_c = 0.8$ ; as  $F_L^{(N)} \rightarrow F_c^+$ , the redundancy diverges logarithmically ( $R_\delta \sim |\ln \varepsilon|$ ), while for  $F_L^{(N)} > F_{th} = 0.874$  the redundancy is identically zero (no Darwinism). (c),(d) Robustness of the tradeoff beyond the analytically solvable limit for  $N = 5$  and  $N = 10$ . Colored points show exact diagonalization results at  $g_X/g_Z = 0.01, 0.05, 0.10$ ; the black curve is the  $g_X = 0$  theorem Eq. (9). The tradeoff remains in excellent agreement with the analytical prediction for  $g_X/g_Z \lesssim 0.05$ , with systematic deviations growing as  $g_X$  increases.

Equation (9) is independent of the microscopic dynamical parameters: the interaction time, coupling strength, and environment size are completely eliminated. The Darwinistic redundancy is therefore determined solely by the post-recovery logical fidelity  $F_L^{(N)}$  and the recovery efficiency  $\eta$ . Since  $\ln \gamma^*(\delta) < 0$ ,  $R_\delta$  decreases monotonically with  $F_L^{(N)}$ , showing that any gain in logical quantum information comes at a quantifiable cost to the redundant encoding of classical information. The tradeoff depends only on the observable logical fidelity, not on the microscopic parameter values that generated it.

Equation (9) also has a direct operational interpretation. Because both the logical fidelity and the Holevo information are determined by the same decoherence factor, syndrome-based recovery and environment-fragment readout provide equivalent experimental probes of the same system–environment information flow. Experimentally, measuring the post-recovery logical fidelity uniquely identifies the underlying decoherence factor. Since the Holevo information of every environment fragment depends on the same quantity, the entire fragment-information curve can then be reconstructed without direct measurements on the environment. Conversely, measurements of the fragment-information curve determine the logical fidelity through Eq. (9). The tradeoff therefore establishes a direct operational bridge between quantum error correction and Quantum Darwinism, demonstrating that logical quantum information preservation and redundant classical information are complementary manifestations of the same underlying system–environment information flow.

**Logarithmic sensitivity near full objectification.**—Defining the fidelity gap  $\varepsilon = F_L^{(N)} - F_c \geq 0$ , which measures the distance from the minimum achievable logical fidelity, with

$F_c = (1 + \eta)/2$ , the tradeoff becomes

$$R_\delta = -\frac{1}{|\ln \gamma^*(\delta)|} \ln(F_L^{(N)} - F_c) + \text{const}, \quad (10)$$

where the additive constant depends only on the recovery efficiency  $\eta$ . Equation (10) shows that the redundancy grows logarithmically as the logical fidelity approaches  $F_c$ , corresponding to complete objectification ( $\Gamma_N \rightarrow 0$ ). Correspondingly, the sensitivity of the redundancy to changes in the logical fidelity is

$$\left| \frac{\partial R_\delta}{\partial F_L^{(N)}} \right| = \frac{1}{|\ln \gamma^*(\delta)| \varepsilon}, \quad (11)$$

which diverges as  $\varepsilon \rightarrow 0^+$ , showing that near complete objectification an arbitrarily small decrease in logical fidelity produces an increasingly large increase in Darwinistic redundancy. For finite environments this growth is naturally cut off by the bound  $R_\delta \leq N$ , while numerical data for different environment sizes collapse onto the exact tradeoff curve (Fig. 1b), confirming that the tradeoff depends only on the logical fidelity and not on the microscopic dynamical parameters. Equation (10) therefore identifies a distinct logarithmic-sensitivity regime in which a small loss of logical quantum information produces a disproportionately large increase in redundant classical records.

**No-go theorem for Darwinistic redundancy.**—We now show that the competition revealed by the tradeoff theorem is not specific to the solvable model, but follows from a general information-theoretic bound.

**Lemma 1** (Block entropy bound). *For any qubit block state  $\rho_B$  with bare logical fidelity  $F_{\text{bare}} = \langle \bar{z}_+ | \rho_B | \bar{z}_+ \rangle$ , the von*

Neumann entropy satisfies

$$S(\rho_B) \leq H_2(F_{\text{bare}}), \quad (12)$$

with equality if and only if  $\rho_B$  is diagonal in the logical basis.

*Proof.* Writing  $\rho_B = \begin{pmatrix} F_{\text{bare}} & c \\ c^* & 1 - F_{\text{bare}} \end{pmatrix}$  in the logical basis, the entropy  $S(\rho_B)$  achieves its maximum value  $H_2(F_{\text{bare}})$  at  $|c| = 0$  (diagonal state) and decreases to zero as  $|c|$  increases to its maximum value  $\sqrt{F_{\text{bare}}(1 - F_{\text{bare}})}$  (pure state). Hence  $S(\rho_B) \leq H_2(F_{\text{bare}})$  for all  $|c|$ . ■

**Theorem 1** (Theorem (No-go)). *Let  $|\Psi\rangle_{BE}$  be a pure state of a qubit block  $B$  and an arbitrary environment  $E$ . If*

$$F_{\text{bare}} > H_2^{-1}[(1 - \delta) \ln 2], \quad (13)$$

*no environment fragment can satisfy the Darwinism criterion, regardless of the Hamiltonian or environment structure. Consequently, Darwinistic redundancy cannot exist ( $R_\delta = 0$ ).*

*Proof.* Suppose, for contradiction, that Darwinistic redundancy exists. Then some environment fragment satisfies  $\chi(F) \geq (1 - \delta) \ln 2$ . Using

$$\chi(F) \leq \chi(E) \leq S(\rho_E) = S(\rho_B)$$

together with Lemma 12, we obtain

$$H_2(F_{\text{bare}}) \geq S(\rho_B) = S(\rho_E) \geq \chi(E) \geq \chi(F) \geq (1 - \delta) \ln 2.$$

Since  $H_2(p)$  is monotonically decreasing on  $p \in [1/2, 1]$ , this implies

$$F_{\text{bare}} \leq H_2^{-1}[(1 - \delta) \ln 2],$$

which contradicts Eq. (13). Therefore no environment fragment can satisfy the Darwinism criterion, and hence Darwinistic redundancy cannot exist. ■

**Corollary.**—For the imperfect-recovery model,  $F_{\text{bare}} = (F_L^{(N)} - \eta)/(1 - \eta)$ , so Eq. (13) becomes

$$F_L^{(N)} > \eta + (1 - \eta) H_2^{-1}[(1 - \delta) \ln 2] \implies R_\delta = 0. \quad (14)$$

For  $\delta = 0.10$  and  $\eta = 0.60$  this threshold is  $F_L^{(N)} > 0.874$ .

*Remark.*—The solvable model saturates every step of the proof chain. The reduced block state is diagonal in the logical basis, giving  $S(\rho_B) = H_2(F_{\text{bare}})$ , while the conditional environment states remain pure, yielding  $\chi(E) = S(\rho_E)$ . Together, these relations collapse the proof chain into the equality

$$\chi(E) = S(\rho_E) = S(\rho_B) = H_2(F_{\text{bare}}).$$

Consequently, every bit of logical entropy generated in the block is converted into accessible classical information in the environment. The solvable model therefore achieves the largest Darwinistic redundancy compatible with a given logical fidelity, while the no-go theorem shows that no other dynamics can exceed this information-theoretic limit.

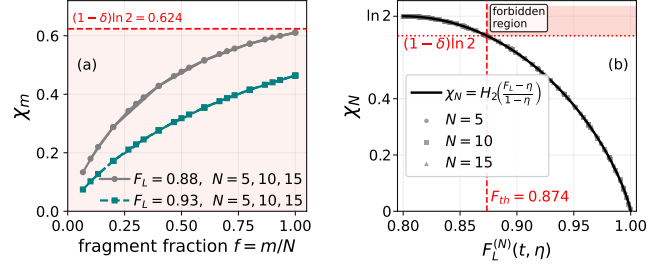


FIG. 2. Numerical confirmation of the no-go theorem,  $\delta = 0.10$ ,  $\eta = 0.60$ . (a) Fragment Holevo information  $\chi_m$  as a function of the fragment fraction  $f = m/N$ , computed for  $N = 5, 10, 15$  at two representative logical fidelities  $F_L = 0.88$  and  $F_L = 0.93$ , both lying above the no-go threshold  $F_{\text{th}} = 0.874$ . Every curve remains strictly below the Darwinism threshold  $(1 - \delta) \ln 2 = 0.624$  nat (red dashed line) for all fragment sizes  $m = 1, \dots, N$ , demonstrating that no environment fragment—however large—carries sufficient classical information to establish Darwinistic redundancy when  $F_L > F_{\text{th}}$ . (b) Full-environment Holevo information  $\chi_N$  as a function of logical fidelity  $F_L^{(N)}$ , for  $N = 5, 10, 15$  at  $g_X = 0$  (closed-form, colored by  $N$ ) and for  $N = 10$  at  $g_X/g_Z = 0.01, 0.05, 0.10$  (exact diagonalization, distinct markers). The black curve is the exact analytical result  $\chi_N = H_2((F_L^{(N)} - \eta)/(1 - \eta))$ . The salmon-shaded rectangle defines the forbidden region:  $F_L > F_{\text{th}}$  and  $\chi_N \geq (1 - \delta) \ln 2$ . Of the data points spanning three environment sizes and four coupling ratios ( $g_X/g_Z = 0-0.10$ ), not a single point enters the forbidden rectangle, confirming that the no-go bound is tight at  $F_L = F_{\text{th}}$  and is never violated.

**Numerical Validation.**—We validate the analytical predictions and test their robustness beyond the analytically solvable limit by exact diagonalization of the full Hamiltonian  $\hat{H} = g_Z \hat{Z}_b \otimes \hat{S}_Z + g_X \hat{X}_b \otimes \hat{S}_X$  for several environment sizes and coupling ratios  $g_X/g_Z$ .

Figure 1a illustrates the Darwinism–QEC competition for  $N = 10$  at  $g_X = 0$ . As the logical fidelity  $F_L^{(N)}$  decreases toward its limiting value, the Darwinistic redundancy  $R_\delta$  grows monotonically, in quantitative agreement with Eqs. (5) and (8).

Figure 1b confirms the logarithmic critical scaling of Eq. (10): plotted as  $R_\delta$  versus  $F_L^{(N)}$  collapse onto the single analytical curve for different  $N$ , verifying that  $N$  and  $t$  enter only through the fidelity gap  $\varepsilon = F_L^{(N)} - F_c$  and demonstrating that the tradeoff is independent of the microscopic dynamical parameters. The redundancy diverges as  $F_L^{(N)} \rightarrow F_c^+$  and vanishes above the no-go threshold  $F_L^{(N)} > 0.874$ , both in excellent agreement with the analytical predictions.

Figures 1c–d demonstrate the robustness of the analytical tradeoff against the non-commuting perturbation  $g_X \hat{X}_b \otimes \hat{S}_X$ . For  $g_X/g_Z = 0.01$ , deviations remain below 0.2%; for  $g_X/g_Z = 0.05$  (0.10) they reach 4% (12%), growing as  $g_X$  approaches the solvable interaction strength.

Figure 2 provides numerical confirmation of the model-independent no-go theorem. Panel (a) shows that, for logical fidelities above the predicted threshold, the Holevo information of every environment fragment remains below the

Darwinism criterion, independent of fragment size. Panel (b) demonstrates that numerical data obtained for different environment sizes and coupling ratios never enter the forbidden region predicted by Theorem 1, confirming the general bound beyond the exactly solvable limit.

**Discussion.**— The present work places the relationship between quantum error correction and Quantum Darwinism on a quantitative footing by establishing a direct information-theoretic bridge between the two. Within an exactly solvable block–environment model, we derive a parameter-free tradeoff between the post-recovery logical fidelity and Darwinistic redundancy. Because both quantities are governed by the same decoherence parameter, the logical fidelity uniquely determines the Holevo information of environment fragments—and hence the Darwinistic redundancy—establishing logical recovery and environment-fragment readout as equivalent operational probes of the same underlying decoherence process. Because the solvable model saturates the underlying information-theoretic bounds, every loss of logical information is exactly balanced by a corresponding gain in classically accessible environmental records.

Beyond the solvable dynamics, the no-go theorem establishes a model-independent information-theoretic constraint on the coexistence of quantum error correction and Quantum Darwinism: the existence of Darwinistic redundancy requires

the bare logical fidelity to lie below a critical threshold, independent of the microscopic Hamiltonian or environment structure. For the imperfect-recovery channel considered here, this general bound translates directly into a threshold for the post-recovery logical fidelity, while the solvable model saturates the bound, demonstrating that it is an extremal realization of the Darwinism–QEC competition. Natural extensions include correlated or initially entangled environments, spatially local interactions, general recovery channels, and higher-dimensional logical encodings. More generally, these results suggest that decoherence, quantum error correction, and the emergence of objective classical information are governed by common information-theoretic principles rather than by the details of a particular microscopic model, providing a unified framework for understanding the quantum-to-classical transition in error-corrected quantum systems.

## I. ACKNOWLEDGEMENT

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### Appendix A: Collective Hamiltonian

The main-text model involves a logical block  $b$  coupled to  $N$  independent environment qubits through the Hamiltonian

$$\hat{H}_N = g_Z \hat{Z}_b \otimes \hat{S}_Z + g_X \hat{X}_b \otimes \hat{S}_X, \quad (\text{A1})$$

where  $\hat{S}_{Z(X)} = \sum_{k=1}^N Z_k(X_k)$  are collective environment spin operators, and  $\hat{Z}_b = Z^{\otimes 3}$ ,  $\hat{X}_b = X^{\otimes 3}$  act on the logical block. The  $g_Z$  interaction produces pure dephasing in the eigenbasis of  $\hat{Z}_b$ , giving rise to logical decoherence in the logical basis, while the  $g_X$  interaction induces coherent logical-state mixing that acts as an uncorrectable logical error channel. The two terms therefore describe the dominant dephasing interaction ( $g_Z$ ) together with a non-commuting perturbation ( $g_X$ ), providing the minimal model used to assess the robustness of the analytical results.

### Appendix B: Many-environment spectrum

For a single environment qubit ( $N = 1$ ),  $\hat{H}_1^2$  is proportional to the identity on the environment, giving the three-level spectrum  $\{0, \pm\sqrt{g_Z^2 + g_X^2}\}$  that underlies the exact closed-form solutions of the main text. For  $N \geq 2$  this simplification no longer holds. A direct calculation using  $\hat{Z}_b^2 = \hat{X}_b^2 = \mathbb{I}_b$  and the Pauli relation  $\hat{Z}_b \hat{X}_b + \hat{X}_b \hat{Z}_b = 0$  gives

$$\hat{H}_N^2 = \mathbb{I}_b \otimes (g_Z^2 \hat{S}_Z^2 + g_X^2 \hat{S}_X^2) - 2g_Z g_X \hat{Y}_b \otimes \hat{S}_Y, \quad (\text{B1})$$

where we used the collective commutator  $[\hat{S}_X, \hat{S}_Z] = -[\hat{S}_Z, \hat{S}_X] = 2i\hat{S}_Y$ . For  $N \geq 2$  the operator  $g_Z^2 \hat{S}_Z^2 + g_X^2 \hat{S}_X^2$  is not proportional to the environment identity; for example, with  $N = 2$ ,

$$g_Z^2 \hat{S}_Z^2 + g_X^2 \hat{S}_X^2 = (g_Z^2 + g_X^2) \cdot 2\mathbb{I} + 2g_Z^2 Z_1 Z_2 + 2g_X^2 X_1 X_2, \quad (\text{B2})$$

which carries a nontrivial two-qubit spectrum. Consequently, closed-form diagonalization is no longer available in general, and exact diagonalization provides the numerical approach used throughout this work.

### Appendix C: Weak-coupling approximation

To obtain analytical progress beyond  $g_X = 0$  we decompose

$$\hat{H}_N = \hat{H}_0 + \hat{V}, \quad \hat{H}_0 = g_Z \hat{Z}_b \otimes \hat{S}_Z, \quad \hat{V} = g_X \hat{X}_b \otimes \hat{S}_X, \quad (\text{C1})$$

and treat  $\hat{V}$  as a perturbation valid for  $g_X/g_Z \ll 1$ . The unperturbed Hamiltonian  $\hat{H}_0$  is exactly solvable for arbitrary  $N$  (main text, Sec. II), providing a systematic starting point for the perturbative analysis which forms the basis of the perturbative analysis presented below.

### Appendix D: Exact dynamics under $\hat{H}_0$

We derive the time-evolved state under the unperturbed Hamiltonian  $\hat{H}_0 = g_Z \hat{Z}_b \otimes \hat{S}_Z$ . The logical block is initialized in the logical codeword  $|\bar{z}_+\rangle_b = (|000\rangle + |111\rangle)/\sqrt{2}$ , corresponding to the encoded logical qubit, while the environment is assumed to begin in an uncorrelated product state with each qubit prepared in  $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ . The initial state is therefore

$$|\Psi(0)\rangle = |\bar{z}_+\rangle_b \otimes |+\rangle^{\otimes N}.$$

Since  $\hat{Z}_b^2 = \mathbb{I}_b$  and  $[Z_i, Z_j] = 0$  for all  $i \neq j$ , the operator powers satisfy  $\hat{H}_0^n = g_Z^n \hat{Z}_b^n \otimes \hat{S}_Z^n$ . Expanding the propagator as a Taylor series and separating even and odd powers, using  $\hat{Z}_b^{2m} |\bar{z}_+\rangle = |\bar{z}_+\rangle$  and  $\hat{Z}_b^{2m+1} |\bar{z}_+\rangle = |\bar{z}_-\rangle$ , gives

$$|\Psi_+(t)\rangle = e^{-it\hat{H}_0} |\Psi(0)\rangle.$$

$$|\Psi_+(t)\rangle = |\bar{z}_+\rangle_b \otimes \sum_{m=0}^{\infty} \frac{(-1)^m (g_Z t)^{2m}}{(2m)!} \hat{S}_Z^{2m} |+\rangle^{\otimes N} - i |\bar{z}_-\rangle_b \otimes \sum_{m=0}^{\infty} \frac{(-1)^m (g_Z t)^{2m+1}}{(2m+1)!} \hat{S}_Z^{2m+1} |+\rangle^{\otimes N}. \quad (\text{D1})$$

The subscript “+” indicates evolution from the initial logical state  $|\bar{z}_+\rangle$ . Recognizing the operator cosine and sine series yields the exact evolved state

$$|\Psi_+(t)\rangle = |\bar{z}_+\rangle_b \otimes \cos(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} - i |\bar{z}_-\rangle_b \otimes \sin(g_Z t \hat{S}_Z) |+\rangle^{\otimes N}, \quad (\text{D2})$$

and the corresponding exact propagator

$$\hat{U}_N(t) = \cos(g_Z t \hat{S}_Z) \mathbb{I}_b - i \hat{Z}_b \otimes \sin(g_Z t \hat{S}_Z). \quad (\text{D3})$$

Since  $\hat{S}_Z = \sum_{k=1}^N Z_k$ , and the operators  $Z_k$  mutually commute, the exponential factorizes as  $e^{-ig_Z t \hat{S}_Z} = \prod_{k=1}^N e^{-ig_Z t Z_k}$ , from which the corresponding factorization of  $\cos(g_Z t \hat{S}_Z)$  and  $\sin(g_Z t \hat{S}_Z)$  follows immediately.

$$\cos(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} = \bigotimes_{k=1}^N \cos(g_Z t Z_k) |+\rangle_k, \quad (\text{D4})$$

and analogously for the sine term. This factorization is the key property that makes the dynamics analytically tractable for arbitrary  $N$ : each environment qubit evolves independently, conditioned on the logical-block state.

## Appendix E: Symmetric evolution of the two logical branches

### 1. Evolution of the $|\bar{z}_+\rangle_b$ branch

Applying the exact propagator Eq. (D3) to the initial state  $|\Psi_+(0)\rangle = |\bar{z}_+\rangle_b \otimes |+\rangle^{\otimes N}$  and using  $\hat{Z}_b |\bar{z}_{\pm}\rangle_b = |\bar{z}_{\mp}\rangle_b$  gives

$$|\Psi_+(t)\rangle = |\bar{z}_+\rangle_b \otimes \cos(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} - i |\bar{z}_-\rangle_b \otimes \sin(g_Z t \hat{S}_Z) |+\rangle^{\otimes N}. \quad (\text{E1})$$

We now introduce the two environment branch states

$$|\phi_+(t)\rangle = e^{-ig_Z t \hat{S}_Z} |+\rangle^{\otimes N}, \quad (\text{E2})$$

$$|\phi_-(t)\rangle = e^{+ig_Z t \hat{S}_Z} |+\rangle^{\otimes N}, \quad (\text{E3})$$

in terms of which the operator cosine and sine act as

$$\cos(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} = \frac{|\phi_+(t)\rangle + |\phi_-(t)\rangle}{2}, \quad (\text{E4})$$

$$\sin(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} = \frac{|\phi_-(t)\rangle - |\phi_+(t)\rangle}{2i}. \quad (\text{E5})$$

Substituting Eqs. (E4)–(E5) into Eq. (E1) and simplifying:

$$|\Psi_+(t)\rangle = \frac{|\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b}{2} |\phi_+(t)\rangle + \frac{|\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b}{2} |\phi_-(t)\rangle. \quad (\text{E6})$$

Note that  $(|\bar{z}_+\rangle_b \pm |\bar{z}_-\rangle_b)/\sqrt{2}$  are proportional to the computational-basis states  $|0\rangle^{\otimes 3}$  and  $|1\rangle^{\otimes 3}$  respectively, so Eq. (E6) expresses the evolved state as a superposition of two definite-parity logical branches, each entangled with a distinct environment state.

### 2. Evolution of the $|\bar{z}_-\rangle_b$ branch

For the orthogonal initial state  $|\Psi_-(0)\rangle = |\bar{z}_-\rangle_b \otimes |+\rangle^{\otimes N}$ , applying  $\hat{U}_N(t)$  and using  $\hat{Z}_b |\bar{z}_{\pm}\rangle_b = |\bar{z}_{\mp}\rangle_b$  gives

$$|\Psi_-(t)\rangle = |\bar{z}_-\rangle_b \otimes \cos(g_Z t \hat{S}_Z) |+\rangle^{\otimes N} - i |\bar{z}_+\rangle_b \otimes \sin(g_Z t \hat{S}_Z) |+\rangle^{\otimes N}. \quad (\text{E7})$$

Substituting the identities Eqs. (E4) and (E5) and simplifying

$$|\Psi_-(t)\rangle = \frac{|\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b}{2} |\phi_+(t)\rangle - \frac{|\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b}{2} |\phi_-(t)\rangle. \quad (\text{E8})$$

a. *Symmetry between the two branches.* Comparing Eqs. (E6) and (E8),

$$|\Psi_+(t)\rangle = \frac{|\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b}{2} |\phi_+(t)\rangle + \frac{|\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b}{2} |\phi_-(t)\rangle, \quad (\text{E9})$$

$$|\Psi_-(t)\rangle = \frac{|\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b}{2} |\phi_+(t)\rangle - \frac{|\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b}{2} |\phi_-(t)\rangle. \quad (\text{E10})$$

The two branches share identical coefficients on  $|\phi_+(t)\rangle$  and differ only in the sign of the  $|\phi_-(t)\rangle$  coefficient. This perfect symmetry reflects the  $\hat{Z}_b \rightarrow -\hat{Z}_b$  invariance of  $\hat{H}_0$ : reversing the logical branch simply reverses the relative phase between the two environment states. The subsequent analysis — decoherence factor, logical fidelity, Holevo information, and Darwinistic redundancy — is governed entirely by the overlap  $\Gamma_N(t) = \langle \phi_-(t) | \phi_+(t) \rangle$ , which is the same for both branches.

## Appendix F: Decoherence factor and logical coherence

The central quantity governing all subsequent results is the overlap between the two environment branch states,

$$\Gamma_N(t) = \langle \phi_-(t) | \phi_+(t) \rangle, \quad (\text{F1})$$

which controls the degree of logical decoherence. As  $\Gamma_N(t)$  decreases toward zero, the environment becomes increasingly capable of distinguishing the two  $\hat{Z}_b$  branches, thereby suppressing their coherence. In the logical basis this loss of coherence appears as population transfer between  $|\bar{z}_+\rangle$  and  $|\bar{z}_-\rangle$ , reducing the logical fidelity while increasing the classical information accessible from the environment. The overlap  $\Gamma_N(t)$  quantifies the distinguishability of the two conditional environment states: smaller overlap corresponds to greater distinguishability and therefore greater information acquired by the environment about the logical block.

### 1. Factorization and single-qubit overlap

Since the operators  $Z_k$  mutually commute,  $e^{\mp i g_Z t \hat{S}_Z} = \prod_{k=1}^N e^{\mp i g_Z t Z_k}$ , and therefore the conditional environment states factorize as  $|\phi_{\pm}(t)\rangle = \bigotimes_{k=1}^N e^{\mp i g_Z t Z_k} |+\rangle_k$ . Consequently, their overlap factorizes as

$$\Gamma_N(t) = \prod_{k=1}^N \langle \phi_-^{(k)}(t) | \phi_+^{(k)}(t) \rangle = [\langle \phi_-^{(1)}(t) | \phi_+^{(1)}(t) \rangle]^N, \quad (\text{F2})$$

where the second equality follows because all environment qubits evolve identically. For a single environment qubit,

$$|\phi_+^{(1)}(t)\rangle = \cos(g_Z t) |+\rangle - i \sin(g_Z t) |-\rangle, \quad (\text{F3})$$

$$|\phi_-^{(1)}(t)\rangle = \cos(g_Z t) |+\rangle + i \sin(g_Z t) |-\rangle, \quad (\text{F4})$$

as  $e^{-i\theta\hat{A}} = \cos\theta \mathbb{I} - i \sin\theta \hat{A}$  for any operator  $\hat{A}$  satisfying  $\hat{A}^2 = \mathbb{I}$ . So the single-qubit inner product is

$$\langle \phi_-^{(1)}(t) | \phi_+^{(1)}(t) \rangle = \cos^2(g_Z t) - \sin^2(g_Z t) = \cos(2g_Z t), \quad (\text{F5})$$

where we used  $\langle \pm | \pm \rangle = 1$ ,  $\langle \pm | \mp \rangle = 0$ . Substituting into Eq. (F2) gives the exact decoherence factor

$$\Gamma_N(t) = \cos^N(2g_Z t), \quad 0 \leq t \leq \frac{\pi}{4g_Z}. \quad (\text{F6})$$

$\Gamma_N(t)$  decays monotonically from  $\Gamma_N(0) = 1$  (no decoherence) to  $\Gamma_N(\pi/4g_Z) = 0$  (complete decoherence), with the decay accelerating as  $N$  increases: each additional environment qubit multiplies the overlap by another factor of  $\cos(2g_Z t) < 1$ , driving  $\Gamma_N$  exponentially toward zero.

### 2. Reduced logical density matrix

Starting from  $|\Psi_+(t)\rangle$  in Eq. (E6), the full density matrix is  $\rho(t) = |\Psi_+(t)\rangle \langle \Psi_+(t)|$ . Introducing the shorthand  $|A\rangle = |\bar{z}_+\rangle_b + |\bar{z}_-\rangle_b$  and  $|B\rangle = |\bar{z}_+\rangle_b - |\bar{z}_-\rangle_b$ , tracing over the environment gives

$$\rho_b(t) = \text{Tr}_E[\rho(t)] = \frac{1}{4} \left[ |A\rangle \langle A| + |B\rangle \langle B| + \Gamma_N(t) (|A\rangle \langle B| + |B\rangle \langle A|) \right], \quad (\text{F7})$$

where we used  $\text{Tr}_E(|\phi_+\rangle\langle\phi_-|) = \text{Tr}_E(|\phi_-\rangle\langle\phi_+|) = \Gamma_N(t)$ , Substituting  $|A\rangle\langle A| + |B\rangle\langle B| = 2(|\bar{z}_+\rangle\langle\bar{z}_+| + |\bar{z}_-\rangle\langle\bar{z}_-|)$  and  $|A\rangle\langle B| + |B\rangle\langle A| = 2(|\bar{z}_+\rangle\langle\bar{z}_+| - |\bar{z}_-\rangle\langle\bar{z}_-|)$  yields the exact reduced block state

$$\rho_b(t) = \frac{1 + \Gamma_N(t)}{2} |\bar{z}_+\rangle\langle\bar{z}_+| + \frac{1 - \Gamma_N(t)}{2} |\bar{z}_-\rangle\langle\bar{z}_-|, \quad (\text{F8})$$

which in the logical basis  $\{|\bar{z}_+\rangle, |\bar{z}_-\rangle\}$  takes the diagonal matrix form

$$\rho_b(t) = \frac{1}{2} \begin{pmatrix} 1 + \Gamma_N(t) & 0 \\ 0 & 1 - \Gamma_N(t) \end{pmatrix}. \quad (\text{F9})$$

The reduced state is diagonal in the logical basis  $\{|\bar{z}_+\rangle, |\bar{z}_-\rangle\}$ , with populations  $p_{\pm}(t) = [1 \pm \Gamma_N(t)]/2$ . Thus, although the interaction is pure dephasing in the  $\hat{Z}_b$  eigenbasis  $\{|000\rangle, |111\rangle\}$ , it appears as population transfer between the logical codewords in the basis used throughout this work. At  $t = 0$ ,  $\Gamma_N = 1$  and  $\rho_b = |\bar{z}_+\rangle\langle\bar{z}_+|$ , whereas  $\Gamma_N \rightarrow 0$  yields the maximally mixed logical state  $\rho_b \rightarrow \mathbb{I}_b/2$ .

### 3. Weak-coupling Gaussian decoherence and timescale

For  $g_Z t \ll 1$ , the Taylor expansion  $\cos(2g_Z t) \approx 1 - 2g_Z^2 t^2$  gives

$$\Gamma_N(t) \approx (1 - 2g_Z^2 t^2)^N \approx e^{-2Ng_Z^2 t^2}, \quad (\text{F10})$$

where the second approximation uses  $(1 - \epsilon)^N \approx e^{-N\epsilon}$  for  $\epsilon = 2g_Z^2 t^2 \ll 1$ . This Gaussian envelope identifies the characteristic decoherence timescale

$$\tau_D = \frac{1}{g_Z \sqrt{2N}}, \quad (\text{F11})$$

at which  $\Gamma_N(\tau_D) = e^{-1}$ . The  $N^{-1/2}$  scaling means that each additional environment qubit accelerates decoherence: adding qubits from  $N$  to  $N + 1$  multiplies  $\Gamma_N$  by a further factor  $\cos(2g_Z t) < 1$ , so even a weakly coupled environment of many qubits can produce extremely rapid dephasing. For example, with  $\cos(2g_Z t) = 0.99$ , one finds  $\Gamma_1 = 0.99$  but  $\Gamma_{1000} = 0.99^{1000} \approx 4 \times 10^{-5}$ : a factor of  $\sim 10^4$  suppression from environment size alone. This collective amplification of decoherence by independent environment qubits is the microscopic mechanism underlying both the degradation of QEC performance and the onset of Darwinistic redundancy studied in the main text.

### Appendix G: Logical fidelity under imperfect recovery

We incorporate imperfect QEC recovery via the channel

$$\mathcal{E}_\eta(\rho) = P_+ \rho P_+ + \eta \hat{Z}_b P_- \rho P_- \hat{Z}_b + (1 - \eta) P_- \rho P_-, \quad (\text{G1})$$

where  $P_{\pm} = |\bar{z}_{\pm}\rangle\langle\bar{z}_{\pm}|$  are the logical projectors and  $\eta \in [0, 1]$  is the recovery efficiency. From the reduced block state Eq. (F8), the syndrome probabilities are

$$p_+(t) = \frac{1 + \Gamma_N(t)}{2}, \quad p_-(t) = \frac{1 - \Gamma_N(t)}{2}. \quad (\text{G2})$$

Since the recovery succeeds with probability  $\eta$  whenever the block occupies the  $|\bar{z}_-\rangle$  sector, the post-recovery logical fidelity is  $F_L^{(N)} = p_+ + \eta p_-$ , which gives

$$F_L^{(N)}(t, \eta) = \frac{1 + \eta + (1 - \eta)\Gamma_N(t)}{2} = \frac{1 + \eta + (1 - \eta)\cos^N(2g_Z t)}{2}. \quad (\text{G3})$$

This closed-form expression is valid for all  $N$ ,  $t$ ,  $\eta$ , and  $g_Z$ . It ranges from  $F_L^{(N)} = 1$  at  $t = 0$  (perfect fidelity, no decoherence) to  $F_L^{(N)} = (1 + \eta)/2 = F_c$  as  $\Gamma_N \rightarrow 0$  (minimum achievable fidelity).

*a. Weak-coupling regime and decoherence timescale.* Using the Gaussian approximation  $\Gamma_N(t) \approx e^{-2Ng_Z^2 t^2}$  derived in Eq. (F10), the fidelity becomes

$$F_L^{(N)}(t, \eta) \approx \frac{1 + \eta + (1 - \eta)e^{-2Ng_Z^2 t^2}}{2}, \quad g_Z t \ll 1. \quad (\text{G4})$$

The fidelity decays on the timescale  $\tau_D$  (Eq. (F11)) already identified from the decoherence factor. The  $N^{-1/2}$  scaling has a direct operational consequence:

$$N \uparrow \implies \tau_D \propto N^{-1/2} \implies F_L^{(N)}(t, \eta) \text{ decays more rapidly}, \quad (\text{G5})$$

Thus, increasing the number of environment qubits simultaneously accelerates the formation of redundant classical records in the environment—the mechanism underlying Quantum Darwinism—and the degradation of the post-recovery logical fidelity, reflecting the corresponding cost to QEC performance. Equation (G3) is the starting point for deriving the exact Darwinism–QEC tradeoff in the following section.

## Appendix H: Holevo information of environment fragments

We now quantify how much classical information about the logical state is accessible to an observer holding  $m$  environment qubits ( $1 \leq m \leq N$ ). This is the key quantity for Quantum Darwinism: redundancy arises when many independent fragments each carry sufficient classical information to identify the logical state of the block.

### 1. Conditional fragment states and their overlap

Since the conditional environment states  $|\phi_{\pm}(t)\rangle$  are product states, the conditional state of the first  $m$  qubits given logical branch  $\pm$  is the pure state

$$|\phi_{\pm}^{(m)}(t)\rangle = \bigotimes_{k=1}^m [\cos(g_Z t)|+\rangle_k \mp i \sin(g_Z t)|-\rangle_k]. \quad (\text{H1})$$

By the same calculation as for the full environment (Sec. F), the overlap of the two conditional  $m$ -qubit states is

$$\gamma_m(t) = \langle \phi_{-}^{(m)}(t) | \phi_{+}^{(m)}(t) \rangle = \cos^m(2g_Z t) = [\Gamma_N(t)]^{m/N}, \quad (\text{H2})$$

where the last equality shows that the  $m$ -qubit fragment overlap is exactly the  $m/N$  power of the full decoherence factor: each environment qubit contributes one factor of  $\cos(2g_Z t)$ .

### 2. Holevo information

With equal prior probabilities  $\frac{1}{2}$  for each logical state, the Holevo information of fragment  $F$  is

$$\chi_m(t) = S(\bar{\rho}_F) - \frac{1}{2}S(\rho_F^{(+)}) - \frac{1}{2}S(\rho_F^{(-)}), \quad (\text{H3})$$

where  $\bar{\rho}_F = \frac{1}{2}(\rho_F^{(+)} + \rho_F^{(-)})$  and  $\rho_F^{(\pm)} = |\phi_{\pm}^{(m)}\rangle\langle\phi_{\pm}^{(m)}|$ . Since both conditional states are pure,  $S(\rho_F^{(\pm)}) = 0$ , so  $\chi_m = S(\bar{\rho}_F)$ .

To find the eigenvalues of  $\bar{\rho}_F$ , define the normalized symmetric and antisymmetric combinations

$$|s\rangle = \frac{|\phi_{+}^{(m)}\rangle + |\phi_{-}^{(m)}\rangle}{\sqrt{2(1 + \gamma_m)}}, \quad |a\rangle = \frac{|\phi_{+}^{(m)}\rangle - |\phi_{-}^{(m)}\rangle}{\sqrt{2(1 - \gamma_m)}}, \quad (\text{H4})$$

which are orthonormal since  $\langle s|a\rangle = 0$ . Acting with  $\bar{\rho}_F$  on  $|s\rangle$ :

$$\bar{\rho}_F |s\rangle = \frac{1}{2}(\langle\phi_{+}^{(m)}|s\rangle |\phi_{+}^{(m)}\rangle + \langle\phi_{-}^{(m)}|s\rangle |\phi_{-}^{(m)}\rangle) = \frac{1 + \gamma_m}{2} |s\rangle, \quad (\text{H5})$$

and similarly  $\bar{\rho}_F |a\rangle = \frac{1 - \gamma_m}{2} |a\rangle$ . The two eigenvalues of  $\bar{\rho}_F$  are therefore

$$\mu_{\pm} = \frac{1 \pm \gamma_m(t)}{2} = \frac{1 \pm \cos^m(2g_Z t)}{2}, \quad (\text{H6})$$

giving the exact Holevo information

$$\chi_m(t) = H_2\left(\frac{1 + \cos^m(2g_Z t)}{2}\right), \quad (\text{H7})$$

where  $H_2(p) = -p \ln p - (1-p) \ln(1-p)$  is the binary entropy.

### 3. Limiting behaviour

At  $t = 0$ :  $\gamma_m = 1$ , so  $\mu_+ = 1$ ,  $\mu_- = 0$ , and  $\chi_m = H_2(1) = 0$  — no information is accessible from any fragment before decoherence begins. At  $t = \pi/4g_Z$ :  $\gamma_m = 0$ , so  $\mu_{\pm} = \frac{1}{2}$  and  $\chi_m = H_2(\frac{1}{2}) = \ln 2$  — the fragment carries one full bit of classical information, perfectly distinguishing the two logical states. For fixed  $t$ ,  $\chi_m$  is strictly increasing in  $m$ : larger fragments are more informative, since  $|\cos^m(2g_Z t)| < |\cos^{m'}(2g_Z t)|$  for  $m > m'$ . This monotonicity is the microscopic basis for the onset of Darwinistic redundancy studied in the following section.

## Appendix I: Darwinistic redundancy

Quantum Darwinism is characterised not merely by information leakage to the environment, but by its *redundant proliferation* into many independent fragments. The redundancy  $R_\delta$  quantifies how many non-overlapping fragments each carry at least a fraction  $(1 - \delta)$  of the maximum classical information  $\ln 2$ .

### 1. Definition and threshold condition

Let  $m_\delta(t)$  be the smallest fragment size satisfying  $\chi_{m_\delta} = (1 - \delta) \ln 2$ . Substituting Eq. (H7):

$$H_2\left(\frac{1 + \cos^{m_\delta}(2g_Z t)}{2}\right) = (1 - \delta) \ln 2. \quad (\text{I1})$$

Since  $H_2$  is monotone decreasing on  $[\frac{1}{2}, 1]$ , this equation has a unique solution in terms of the *critical overlap*

$$\gamma^*(\delta) = 2H_2^{-1}[(1 - \delta) \ln 2] - 1, \quad (\text{I2})$$

where  $H_2^{-1}$  is the upper-branch inverse ( $p \in (\frac{1}{2}, 1)$ ). Equation (I1) then becomes  $\cos^{m_\delta}(2g_Z t) = \gamma^*(\delta)$ , with  $\gamma^* \in (0, 1)$  depending only on  $\delta$ . The Darwinistic redundancy is

$$R_\delta(t) = \frac{N}{m_\delta(t)}. \quad (\text{I3})$$

### 2. Closed-form redundancy

Taking the logarithm of  $\cos^{m_\delta}(2g_Z t) = \gamma^*$  and solving for  $m_\delta$  gives

$$m_\delta(t) = \frac{\ln \gamma^*(\delta)}{\ln[\cos(2g_Z t)]}, \quad (\text{I4})$$

where both numerator and denominator are negative for  $t \in (0, \pi/4g_Z)$ , ensuring  $m_\delta > 0$ . Substituting into Eq. (I3) yields the exact result

$$R_\delta(t) = N \frac{\ln[\cos(2g_Z t)]}{\ln \gamma^*(\delta)}, \quad t \in \left(0, \frac{\pi}{4g_Z}\right). \quad (\text{I5})$$

Both logarithms are negative so  $R_\delta > 0$ . As  $t \rightarrow 0$ ,  $R_\delta \rightarrow 0$  (no coupling, no redundancy); as  $t \rightarrow \pi/4g_Z$ ,  $\cos(2g_Z t) \rightarrow 0$  and  $R_\delta \rightarrow +\infty$  (maximum decoherence, unbounded redundancy for continuous  $m_\delta$ , cut off at  $N$  for finite environments).

### 3. Weak-coupling limit

For  $g_Z t \ll 1$ , the expansion  $\ln[\cos(2g_Z t)] \simeq -2g_Z^2 t^2$  gives

$$m_\delta(t) \simeq \frac{C(\delta)}{g_Z^2 t^2}, \quad R_\delta(t) \simeq \frac{N g_Z^2 t^2}{C(\delta)}, \quad (\text{I6})$$

where  $C(\delta) = \frac{-\ln \gamma^*(\delta)}{2}$  is a positive,  $\delta$ -dependent constant. The scaling  $R_\delta \propto N g_Z^2 t^2$  shows that redundancy grows linearly in  $N$  (more environment qubits provide more independent copies of the logical-state information) and quadratically in the interaction parameter  $g_Z t$  (stronger coupling imprints more information per qubit). The same parameter  $g_Z t$  drives both the increase in redundancy and the decrease in logical fidelity (Eq. (G3)), making the Darwinism–QEC competition directly visible at the level of the microscopic coupling.

### Appendix J: Exact Darwinism–QEC tradeoff

The logical fidelity Eq. (G3) and the redundancy Eq. (I5) both depend on a single microscopic parameter: the decoherence factor  $\Gamma_N(t) = \cos^N(2g_Z t)$ . Eliminating  $\Gamma_N$  between them yields a direct, parameter-free relation.

From Eq. (G3),

$$\Gamma_N(t) = \frac{2F_L^{(N)} - (1 + \eta)}{1 - \eta}, \quad (\text{J1})$$

so

$$\ln[\cos(2g_Z t)] = \frac{1}{N} \ln \left[ \frac{2F_L^{(N)} - (1 + \eta)}{1 - \eta} \right]. \quad (\text{J2})$$

Substituting into Eq. (I5), the  $N$  in the numerator and the  $1/N$  in Eq. (J2) cancel exactly, giving

$$R_\delta = \frac{1}{\ln \gamma^*(\delta)} \ln \left[ \frac{2F_L^{(N)} - (1 + \eta)}{1 - \eta} \right]. \quad (\text{J3})$$

This is the central result: Eq. (J3) is exact, contains no approximation, and is independent of  $t$ ,  $g_Z$ , and  $N$ . The cancellation of  $N$  reflects the fact that both  $F_L^{(N)}$  and  $R_\delta$  are determined by  $\Gamma_N$  alone: knowing the post-recovery fidelity uniquely fixes the decoherence factor and hence the entire Darwinistic information structure of the environment.

*a. Non-negativity.* Since  $F_L^{(N)} \in [(1 + \eta)/2, 1]$ , the argument of the logarithm lies in  $(0, 1]$ , so  $\ln[\dots] \leq 0$ . Combined with  $\ln \gamma^* < 0$ , this gives  $R_\delta \geq 0$ .

*b. Strict monotone competition.* Differentiating Eq. (J3) with respect to  $F_L^{(N)}$ ,

$$\frac{dR_\delta}{dF_L^{(N)}} = \frac{1}{\ln \gamma^*(\delta)} \cdot \frac{2}{2F_L^{(N)} - (1 + \eta)} < 0, \quad (\text{J4})$$

since both factors are negative. Redundancy is therefore a strictly decreasing function of logical fidelity: every improvement in QEC performance reduces the Darwinistic redundancy and vice versa.

*c. Boundary values.* At  $F_L^{(N)} = 1$  (perfect QEC): the argument equals  $(1 - \eta)/(1 - \eta) = 1$ , so  $\ln(\dots) = 0$  and  $R_\delta = 0$  — no classical records proliferate in the environment. As  $F_L^{(N)} \rightarrow F_c^+$  where  $F_c = (1 + \eta)/2$ : the argument  $\rightarrow 0^+$ , so  $\ln(\dots) \rightarrow -\infty$  and  $R_\delta \rightarrow +\infty$  — unbounded redundancy at maximum decoherence, cut off at  $R_\delta = N$  for finite environments.

#### 1. Critical scaling near $F_c$

Writing  $\varepsilon = F_L^{(N)} - F_c$ , Eq. (J3) becomes

$$R_\delta = \frac{1}{\ln \gamma^*(\delta)} \ln \left[ \frac{2\varepsilon}{1 - \eta} \right] = -\frac{1}{|\ln \gamma^*(\delta)|} \ln \varepsilon + \text{const}, \quad (\text{J5})$$

where  $\text{const} = -\ln[2/(1-\eta)]/|\ln \gamma^*|$  depends only on  $\eta$  and  $\delta$ . As  $\varepsilon \rightarrow 0^+$ ,  $R_\delta$  diverges logarithmically,

$$R_\delta \sim \frac{|\ln \varepsilon|}{|\ln \gamma^*(\delta)|} \quad \text{as } F_L^{(N)} \rightarrow F_c^+, \quad (\text{J6})$$

and accordingly,

$$\left| \frac{dR_\delta}{dF_L^{(N)}} \right| = \frac{1}{|\ln \gamma^*(\delta)| \varepsilon} \sim \varepsilon^{-1} \quad (\text{J7})$$

diverges as a power law, quantifying the critical sensitivity of the Darwinism–QEC competition near the onset of complete decoherence.

### Appendix K: Logical fidelity determines the Darwinism spectrum

The exact tradeoff Eq. (J3) relates the logical fidelity to the Darwinistic redundancy  $R_\delta$ , which depends only on the threshold-crossing fragment size  $m_\delta$ . A strictly stronger result holds: the logical fidelity uniquely determines the Holevo information  $\chi_m$  of every environment fragment, for all  $m = 1, \dots, N$ .

Within the present model, eliminating  $\Gamma_N(t)$  from Eq. (H7) using Eq. (J1) and the identity  $\cos^m(2g_Z t) = \Gamma_N^{m/N}(t)$  (valid for  $t \in [0, \pi/4g_Z]$ ) yields the exact result:

For every fragment of  $m$  environment qubits,

$$\chi_m = H_2 \left[ \frac{1}{2} \left( 1 + \left( \frac{2F_L^{(N)} - (1+\eta)}{1-\eta} \right)^{m/N} \right) \right]. \quad (\text{K1})$$

Equation (K1) contains no explicit dependence on the interaction time or coupling strength. Once the logical fidelity and the fragment fraction  $m/N$  are specified, the Holevo information of every environment fragment is uniquely determined without reference to the microscopic dynamics. This is strictly stronger than the redundancy tradeoff, which determines only the threshold crossing at  $m = m_\delta$ ; here the entire partial-information curve  $\chi_m$  for all  $m$  is fixed by  $F_L^{(N)}$  alone.

Since  $\chi_m = H_2 \left[ \frac{1+\Gamma_N^{m/N}}{2} \right]$ , and  $H_2((1+x)/2)$  is monotonically decreasing on  $x \in [0, 1]$ , the Holevo information increases monotonically with fragment size,  $\chi_1 \leq \chi_2 \leq \dots \leq \chi_N$ . Thus progressively larger environment fragments reveal increasingly more classical information about the logical state, with the full environment attaining the maximum accessible information.

#### 1. Full-environment Holevo information

For the complete environment ( $m = N$ ), the exponent  $m/N = 1$  and Eq. (K1) simplifies to

$$\chi_N = H_2 \left[ \frac{1}{2} \left( 1 + \frac{2F_L^{(N)} - (1+\eta)}{1-\eta} \right) \right] = H_2 \left( \frac{F_L^{(N)} - \eta}{1-\eta} \right), \quad (\text{K2})$$

Equation (K2) establishes a direct observable-to-observable connection: measuring the post-recovery logical fidelity immediately fixes the total classical information encoded in the full environment. Together with Eq. (K1), this shows that a measurement of the post-recovery logical fidelity reconstructs the complete Holevo-information curve  $\{\chi_m\}_{m=1}^N$ , establishing logical-fidelity measurements and environment-fragment readout as operationally equivalent probes of the same decoherence process.

### Appendix L: Exact information balance

The exact solution reveals a precise relationship between the entropy generated in the logical block and the classical information acquired by the complete environment.

From Eq. (F8), the reduced block state is diagonal in the logical basis with eigenvalues  $\lambda_{\pm} = (1 \pm \Gamma_N)/2$ , giving the von Neumann entropy

$$S(\rho_b) = H_2\left(\frac{1 + \Gamma_N(t)}{2}\right). \quad (\text{L1})$$

The full-environment Holevo information (Eq. (H7) with  $m = N$ ) is

$$\chi_N = H_2\left(\frac{1 + \Gamma_N(t)}{2}\right). \quad (\text{L2})$$

Comparing Eqs. (L1) and (L2) gives the exact information balance

$$S(\rho_b) = \chi_N. \quad (\text{L3})$$

Every nat of von Neumann entropy generated in the logical block is exactly accounted for by the Holevo information stored in the full environment — no information is hidden in quantum coherences between the conditional environment states.

This equality holds because the two contributing factors are both determined by  $\Gamma_N(t)$  through the same function  $H_2((1 + \Gamma_N)/2)$ . Microscopically, the two conditions responsible are: (i) the conditional environment states  $|\phi_{\pm}(t)\rangle$  are pure (product states), so  $S(\rho_E^{(\pm)}) = 0$  and  $\chi_N = S(\bar{\rho}_E)$ ; and (ii)  $S(\bar{\rho}_E) = S(\rho_b)$ , which follows from the eigenvalue structure shared by  $\bar{\rho}_E$  and  $\rho_b$  through the common decoherence factor  $\Gamma_N$ .

### 1. Connection with logical fidelity

Substituting  $\Gamma_N = (2F_L^{(N)} - (1 + \eta))/(1 - \eta)$  from Eq. (J1), both sides of Eq. (L3) become

$$S(\rho_b) = \chi_N = H_2\left(\frac{F_L^{(N)} - \eta}{1 - \eta}\right). \quad (\text{L4})$$

The logical fidelity therefore simultaneously determines the block entropy, the full-environment classical information, and — through Eq. (K1) — the Holevo information of every environment fragment. All three are governed by the single decoherence parameter  $\Gamma_N(t)$ , which is in turn uniquely fixed by the observable  $F_L^{(N)}$ .

## Appendix M: General fragment-information bound

Before proving the model-independent no-go theorem, we derive a general upper bound on the classical information accessible from an arbitrary environment fragment. Unlike the exact relations obtained for the solvable model, the following bound relies only on general information-theoretic properties and the purity of the joint block–environment state.

*a. Model-independent part.* For any bipartite state and any subsystem  $F \subseteq E$ , the partial trace  $\mathcal{T}_F = \text{Tr}_{E \setminus F}$  is a completely positive trace-preserving (CPTP) map. The data-processing inequality for Holevo information gives  $\chi(F) \leq \chi(E)$ . The Holevo bound further gives  $\chi(E) \leq S(\rho_E)$ , and for a pure joint state  $|\Psi\rangle_{bE}$  the purification identity gives  $S(\rho_E) = S(\rho_b)$ . Chaining these three inequalities yields the *model-independent* bound

$$\chi(F) \leq \chi(E) \leq S(\rho_b), \quad (\text{M1})$$

which holds for any Hamiltonian, any  $N$ , and any initial state, requiring only that the joint block–environment state is pure.

*b. Model-specific saturation.* In the present solvable model, the conditional environment states are pure, so  $\chi(E) = S(\bar{\rho}_E)$ . Since  $\bar{\rho}_E$  and  $\rho_b$  share the same nonzero eigenvalue spectrum,  $\chi(E) = S(\rho_b)$ , showing that the general upper bound is saturated.

For fragments consisting of  $m$  environment qubits, Eq. (M1) becomes

$$\chi_m \leq S(\rho_b).$$

Combining this general inequality with the entropy bound  $S(\rho_b) \leq H_2(F_{\text{bare}})$  proved in the main text immediately yields

$$\chi_m \leq S(\rho_b) \leq H_2(F_{\text{bare}}),$$

which forms the key ingredient of the model-independent no-go theorem.